Angular Hecke 7/8 investigation summary

The exact Eisenstein-family theorem, its proof certificate, and a separate reproducibility defect.

Assessment: The checked public proof passes. A separate repaired copy now passes a full clean source build and fresh native kernel replay. The original packet retains its reproducibility defect; the theorem and mathematical argument are unchanged.

Criterion	Verdict
Mathematical soundness	**ACCEPT**
Statement and scope fidelity	**ACCEPT**
Imported input applicability	**ACCEPT**
Coefficients, normalization and uniformity	**ACCEPT**
Verification evidence and reproducibility	**ACCEPT (repaired)**

The theorem audited

Over K = Q(sqrt(-3)), with its Eisenstein integer ring, the coefficient on (a) is nu(a) (a/|a|)^k. The integer k and finite residue character obey unit compatibility. The prescribed modulus-prime zeros are retained.

The conclusion covers all compatible characters, both angular signs, all nonzero ideal moduli, and primitive and imprimitive presentations. It asserts nonvanishing for **Re(s) > 7/8**. The only exception is s = 1 for angular type zero and principal residue character. The boundary line itself is not claimed.

All real unitary norm twists are included: N(I)^(+i tau) gives L(s-i tau), with principal pole 1+i tau. The brief's negative-sign convention gives L(s+i tau), with pole 1-i tau. No estimate uniform over unbounded angular order is asserted.

Verdict 5 is ACCEPT for the separately repaired copy and remains REJECT for the original ZIP. The repair and new independent evidence are detailed on pages 4-5.

Baseline independent checks

The critical mathematical chain

The audit traced the LOW and FULL energy routes, the counted normalized integral and floor branch, the actual zero witness and natural-moment induction, source correction and common-signal Mellin identities, and finite-order and physical cusp transformations.

The final argument selects a positive improvement margin before the target character. Low-probe growth and the common-signal error then move the continuation boundary below the fixed-cap zero supremum. No attainment of that supremum is assumed. Exact finite Euler deletion restores the original function.

The inspected estimates retain common Fourier densities, the 1/sqrt(NP) slot coefficient, squared counting factors, inverse-normalizer losses, and the paired repeated-prime/conductor allocation. The principal residues contribute R and R/6. A source scalar is divided out only after its nonvanishing is proved.

Proof-object and semantic checks

• **129,614 constants:** the actual public theorem declaration closure was traversed and replayed in a fresh native Lean kernel environment, with no missing dependencies.

• **Three standard axioms:** propext, Classical.choice and Quot.sound. Their types were checked separately. The closure contains no unsafe or partial declarations that the replay would skip.

• **Nine semantic/series modules rebuilt:** seven serialized objects match byte for byte. A separate comparison found all 205 introduced declaration types and bodies identical across all nine modules, including the two differing object files.

• **Exact arithmetic:** 24 rational exponent certificates and three denominator-cleared correction identities passed independent checks.

Integrity and compilation evidence

Independent checks covered 10,688 payload hashes, 8,229 indexed source/import entries and 1,183 angular receipt/source associations. Three public roots were recompiled with cached imports. A separate bounded clean build reached 12 dependency modules; it was not a complete source build.

The author's historical records report 263 incremental accepted modules, 638 selected axiom outputs and 1,052 artifact checks. Those are supplied records, not independent audit counts. The package itself disclaims a full clean build and omits Lean and compiled caches.

Baseline trust boundary: the successful replay used locally available author-cache dependencies and freshly compiled roots under Lean 4.34.1. It checked the proof objects rather than trusting their compilation history. It was not a different kernel implementation or a full source-to-object reconstruction. The new clean build on page 5 addresses the source reconstruction gap.

The reproducibility defect

Earliest observed failure

lean/AngularSieveDensity.lean:21 references the unavailable predicate OAI.CubicEisenstein.AngularSieveModels.HasCanonicalThetaModels. Line 34 calls the unavailable AngularShortEnergy.canonicalShortState_uniform_twist. Independent re-elaboration reproduced these errors.

The historical successful compilation used an older AngularShortEnergy object. The final-root configuration selects a different object. The staged hashes match the selected original files, so this is not a staging corruption.

The first 600-second replay timeout was inconclusive. A later unrestricted replay completed with a missing-constant error. A separate replay of the public theorem's actual proof dependencies passed: the obsolete declarations are outside that closure.

Consequences and minimum repair

The historical module records cannot certify one coherent current source build. This failure does not invalidate the checked public proof or demonstrate that the mathematical theorem is false.

The minimum repair is finite source/build reconciliation: remove an unused obsolete import route while preserving theorem statements and proof bodies, recompute the retained closure, build it from fresh objects, and replay that result. This repair and its completed checks are recorded on pages 4-5.

Limits and source map

The audit did not manually reread every lower helper or foundation, all original Kubota/Patterson papers, or both background manuscripts in full. Complete source/object correspondence, a portable clean build, an official binary-byte comparison and an external-kernel check were not established in the baseline investigation.

Principal packet locations: TARGET_THEOREM.md; AngularCharacter:48-173; AngularFamilyContinuation:16-32; AngularPhysicalFinalAssembly:36-120; AngularPhysicalNormTwistNonvanishing:13-48; AngularPhysicalDetectorActualCountedDyad:58-93; Detector/RowRadicalConductor:318-339; AngularSieveDensity:21,34. OAI paths are under lean/OAI/NumberTheory/DirichletL/.

Full reasoning and receipts: ANGULAR_HECKE_7_8_FINAL_AUDIT.txt and ANGULAR_HECKE_7_8_FINAL_AUDIT_EVIDENCE.zip, alongside this PDF. The evidence includes the native replay, semantic comparison, axiom census, exact arithmetic and source-read hashes.

Original packet SHA-256
ebb393402ba54b9c86c1f1ec13ac551085ae136d04576660641f17e7537a7db0

Repair and updated verdict 5

ACCEPT for the repaired public theorem closure. The original packet remains REJECT on reproducibility. Verdicts 1-4 remain ACCEPT.

Minimum source repair

AngularNormTwistContinuation mixed the required shift identities with an unused older 23/24 seed corollary. That corollary, physical_seed_all_norm_twists, was moved verbatim into AngularNormTwistSeedContinuation together with its seed import. Required declarations, the public theorem, its quantifiers and its proof are unchanged.

Exact source comparison found 8,228 original Lean files byte-identical. The one other original file contains the unchanged retained declarations; the moved statement and proof are preserved verbatim in the new legacy file. Sixteen obsolete modules leave the public import closure and remain archived unchanged.

Failure caught in the repair itself

The first auditor repair script assumed CRLF line endings. It moved the unused corollary but left the LF-terminated seed import in place. The first preservation check repeated that mistake. The full clean build caught the resulting missing-module error after 8,211 successful compilations; the earlier cached-import diagnostic had not caught it.

The import was then removed with an explicit line-content check. A new independent audit uses Lean's own header parser to compare every source import against the index. All 8,213 headers agree. The failed attempt is preserved and distinguished from the final successful result.

Checked continuation of the clean build

The 8,211 successful compilations came from this investigation's fresh object tree. Their unchanged source/import records, compiler and recipe, dependency fingerprints, artifacts and logs were rechecked before reuse. Explicit reuse receipts link to the original compilation records. The corrected continuation module and the remaining public root were then compiled.

The late evidence copy captured corrected versions of some explanatory files. The failed source and index snapshots were reconstructed to the exact hashes in the original compiler receipt and build identity. Raw compiler receipts, logs and results were preserved unchanged. This provenance is recorded explicitly.

Clean-build and replay evidence

Full source build: all 8,213 retained non-toolchain modules compiled successfully, including both original public roots. Imports came only from fresh build outputs and the pinned Lean 4.34.1 toolchain. No author cached objects or historical setup files were admitted.

Artifact verification: 20,519 compiled artifact parts, all source hashes, source/import headers, direct dependency fingerprints, compilation logs and receipts passed checks before replay.

Fresh kernel replay: a harness importing both public roots added no declarations. Lean checker replayed the combined imported environment in a fresh kernel environment and exited successfully. The public proof census found no missing, unsafe or partial dependencies, and only propext, Classical.choice and Quot.sound; their standard types were checked again.

Semantic preservation: all eight retained continuation declarations match the audited baseline in type, definition body and proof body. This supplemental comparison is separate from the clean build and fresh replay.

Evidence and remaining limits

The repaired ZIP includes the exact source patch, source-preservation checks, build commands, per-module receipts and logs, hash-checked metadata, replay records and reproduction scripts. Compiled objects and the runtime are omitted; the complete retained sources are included.

This is a native Lean kernel check on the recorded Windows toolchain. An external kernel implementation, a second operating system and an official binary-byte comparison were not checked. The 16 excluded legacy modules and the relocated legacy corollary are preserved but are not certified as separately buildable. The original audit's limits on manual rereading of lower helpers, papers and background manuscripts remain.

Repair classification: finite module/build reconciliation. No new analytic estimate or change to the mathematical argument was required.